

HW4 Programming Report

Name: Jiaqing Guan

1. Implementation

In this homework, I coded logistic regression and the perceptron. For logistic regression, I used the sigmoid function and gradient descent to update the weights. For the perceptron, I updated the weights only when the prediction was wrong. I tested both methods on the simple dataset and the starplus brain data, and the results matched the expected performance described in the assignment.

2. Results

(1) Logistic regression, simple data

Command: `python3 Linear_Classifiers.py --data simple --algorithm logistic`

Accuracy: training set: 1.0

Accuracy: test set: 1.0

(2) Logistic regression, starplus data

Command: `python3 Linear_Classifiers.py --data starplus --algorithm logistic`

Accuracy: training set: 1.0

Accuracy: test set: 0.7142857142857143

(3) Perceptron, simple data

Command: `python3 Linear_Classifiers.py --data simple --algorithm perceptron`

Accuracy: training set: 1.0

Accuracy: test set: 1.0

(4) Perceptron, starplus data

Command: `python3 Linear_Classifiers.py --data starplus --algorithm perceptron`

Accuracy: training set: 1.0

Accuracy: test set: 0.7857142857142857